

Robert Fielding

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Backend programmer with 30 years of hands-on technical experience

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Experience

ZetaBravo

AI Developer

May 2025–July 2025

Developed an AI application in Python and React, primarily as a backend developer. Used LangGraph and an in-house Ollama server. Focused on performance measurement, bug fixes, and testing for a tool-calling application supporting general-purpose LLMs.

Great Harvest

Bakery Team Member

February 2025–March 2025

Worked in a bakery making bread while pursuing the next technical role.

Leidos

Computer Programmer

September 2023–October 2024

Worked on ProSight, a successful airport physical-security project. Integrated body scanners and explosives-and-drugs detection equipment into the ProSight network, with additional work on carry-on baggage scanners. Implemented Keycloak integration using C# and Go. Traveled on-site to reverse-engineer device protocols, including HTTPS traffic capture through a proxy. Also generated tabular data to train models with PyTorch and Python.

Technologies: C# (9 months); Go (9 months); Python/PyTorch (3 months); cryptography/mTLS (4 months)

greymatter.io — Alexandria, VA

Computer Programmer

October 2015–July 2023

Joined a hybrid commercial and government startup at its inception and introduced Go to the company. Built a production file-upload service used for hundreds of terabytes of data over many years. The system provided high-performance audio, video, and document attachments with enforced security labels, using Go, S3, MySQL, NGINX, Gatekeeper, and mTLS.

Worked on successors supporting government and commercial use cases, prototyped in Python and built with Go, S3, and MongoDB. Used Amazon Rekognition to identify faces, label content, detect inappropriate material, and navigate among familiar faces. Indexed results, including machine-learning labels, in Elasticsearch. Automated GPU-based reinforcement-learning tasks for intern teams.

Technologies: Go (8 years); Java (4 years, intermittently maintaining dependencies); MySQL (4 years); MongoDB (4 years); Docker (8 years); Linux (8 years); cryptography/mTLS (6 years)

Wizdom Music

Self-Employed iPhone/iPad Developer

October 2010–October 2012

Released Mugician one month after the iPad launched. The app saw significant real-world use by well-known musicians, including members of Underoath, Dream Theater, Gorillaz, and The Flying Eyes. Collaborated with other iOS developers on MIDI implementations for the unusual capabilities of a continuous multi-touch surface.

Technologies: Signal processing (2 years); C, C++, and Objective-C (2 years); iPhone development (2 years)

NFR Security / Check Point Software

Computer Programmer

October 2004–October 2015

NFR was an early IPS/IDS pioneer whose nCode language inspected network

Skills

Go, Java, Python, and C

Performance testing, build automation, security analysis, and cryptography

Languages & Tools

Go, Java, Python, C, Bash, C#, JavaScript, OpenSCAD (3D printing), Arduino (embedded C++), Erlang, Lean 4, and TLA+

Award

2015 Application of the Year

Defense Intelligence Agency

Recognized for work on Object Drive, a Dropbox-like application for CAPCO/NATO-classified information with full-text search.

traffic in real time. Helped build the Java desktop Alert Browser and a heavily customized J2EE application server based on JBoss with a MySQL backend. The Sentivist Alert Browser handled millions of events per customer per day and provided intuitive, real-time visualizations of alert streams.

Daily work included network analysis, reading packet captures, decoding HTTP, packet reassembly, and automating network setup and firewall rules. Automated a multi-machine build that checked out source code and produced an ISO with Windows and Linux installers.

Built a debugger for nCode (later renamed cpCode) using WFC, a Windows C# UI framework. The C# interface invoked a Cygwin-built runtime for the C-based cpCode engine. Also worked on the engine as it was incorporated into a Linux kernel module for IPS functionality.

Contributed for about six months to the Check Point Ext JS web console. Worked on SensorNet, an experimental Hadoop analysis project applying PageRank to graphs of communicating nodes. Also worked on Project MSSP, a web-based log collector for subscribed customers' Check Point firewalls.

Technologies: C and C++ (4 years); Ext JS and PHP (2 years); Java (6 years); Python, Celery, and RabbitMQ (2 years); Linux (10 years); cryptography (8 years)

ComTek

October 1999–October 2004

Software Developer

Worked on-site at DISA and Northrop Grumman (formerly Logicon), building software deployments. Taught DISA employees security and cryptography concepts, including X.509 certificates, hashes, DES ciphers, key handling, and inverses. Primarily developed an LDAP-based user and attribute management system. Built DoD PKI websites backed by virtual databases for DISA.

Technologies: Java (4 years); ColdFusion (4 years); Linux (2 years); SunOS (2 years)

Federal Reserve

1996–1999

Software Developer

Developed banking-surveillance applications, primarily in PowerBuilder with occasional Win32 C API work. Introduced Java for these applications; the first production Java deployment used SilverStream.

Technologies: Java (2 years); PowerBuilder (2 years); C/Win32 (1 year)

Education

George Mason University — Fairfax, VA

1993–1996

B.S. in Computer Science

Focused on compilers, operating systems, discrete mathematics, and computer graphics before OpenGL libraries were available on Windows. Programmed in C, Java, and C++ and analyzed compilers and operating systems.

Technologies: Java (2 years, beginning with HotJava Alpha); C (4 years); C++ (2 years); Linux (2 years)